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POUGHKEEPSIE

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Dept. 539, Bldg. 965
May 19, 1958

MEMO TO: Mr. S. W. Dunwell

SUBJECT: 7000A Performance Comparison to 704-709

The Project 7000 Planning Group has recently programmed a number of short problems which begin to give us an insight into the value to technical customers of some of the new features of the 7000 Series.

The problems are discussed in separate memos. They are summarized on the following page in terms of the speed ratio of the 7000A, compared to the 704 or 709. New figures for some earlier problems are also shown to incorporate the current thinking of the design group as to measures for improving the 7000A performance-to-cost ratio. The figures are not final since the design is subject to further modifications until July 1, 1958, as scheduled.

These are all problems of significance to 704-709 customers. In many of the problems, the speed ratio per instruction is only 2 or 3. The large ratios are almost entirely due to new features not available on the 704 and 709, such as variable field length and bit manipulation, or to data processing techniques, such as sorting, which 704-709 customers have neglected to exploit.

The figures give ample support to the 7000 Series philosophy of giving both technical and business customers in a single machine the best techniques from both fields of application.

W. Buchholz
Manager
Project 7000 Planning

WB/mms
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Problem		Speed Ratio
1.	Mesh Calculation	7.7
2.	Monte Carlo	8.6
3.	Composite Test Problem	7.0
4.	Multiplication of 64th order Matrices with 90% Zero Elements	
	a) Single precision on both	39*
	b) Double precision on 709 only	81*
	c) Double precision on both	50*
5.	SAC Problem #1	
	a) Using logic of 704 operational program	8
	b) Using sorting techniques on 7000A	350
6.	SAC Problem #2	10
7.	Card Conversion	18
8.	Print Editing	37

* These figures are compared to the 709. The 704 would take considerably more time.

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